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Description - This material covers the theory of operation of the Transient Noise Filter (TNF).

1- TRANSIENT NOISE FILTER THEORY

The Transient Noise Filter assembly detects and eliminates the presence of spike noise from each of the RF data channels and provides a means of counting and displaying each incidence of spike noise removal during the time of data acquisition for each channel collectively.

The TNF also provides a means of alert and shutdown of the systems cabinet due to any overtemperature condition in the cabinet.

One Transient Noise Filter Expansion (TNFE) Module is added to the TNF assembly in support of the phased array option.

An RS232 communication interface for remote monitoring of detected spike incidences is also provided.

Throughout this section, reference is made to the center operating frequency F_0, which is defined as the following, depending on the system magnetic field strength:

1.5 T → F_0 = 63.86 mHz
1.0T → F_0 = 42.57 mHz

The passband to the RF received signal for installed base systems is flat to within 0.05 dB over a 50-kHz sliding window over the following frequency ranges:

1.5T → F_0 ±500 kHz
1.0T → F_0 ±150 kHz

The pass band to the RF received signal for EPI systems for all field strengths is flat to within 0.5db over a bandwidth of F_0 ±800 kHz.

1-1 Impulse Noise Filter

The TNF and TNFE modules provide an Impulse Noise Filter (INF) function for each RF receiver channel.

1-2 INF MR Signal Delay

The delay function delays the MR signal to match the delay inherent in the detector circuitry for proper placement of the switch blanking window relative to the presence of the spike noise.

1-3 INF Spike Detection Band

The detector has sufficient bandwidth to detect the spike energy outside of the normal MR information signal band but within the band of the received signal preamp.

1-4 INF Spike Noise Detection Threshold

For installed base systems, the spike detection threshold is set such that, when the TNF is enabled, no spike noise that exceeds six times the peak noise is observed at the TPS/ISE receiver A/D input.

For Signa systems, the spike detection threshold is set such that, when the TNF is enabled, no spike noise that exceeds two times the peak noise is observed at the TPS receiver A/D input, or six times the peak noise when observed at the fast receiver quadrature A/D inputs.

1-5 INF Spike Detected Signal

Upon detection of spike noise, each INF detector provides a spike-detected signal to the blanker switch trigger. The spike-detected signal, at a minimum, equals the duration of the spike noise, but does not exceed the duration of the spike by more than one microsecond.

1-6 INF Spurious Coherent Noise Rejection

The detector is designed to reject, or be unresponsive to, any spurious coherent noise that may normally be present outside the MR signal band, but within the passband of the RF Preamp. The TNS is not required to reject coherent noise that is present due to a break, or leakage, in the normal RF integrity of the scanner screen room, or a break in the normal shield integrity of any other GE MRI scanner equipment contained within the screen room. The TNS is not required to reject coherent noise present due to RF leakage from other nonGE MRI scanner equipment brought into the screen room.

1-7 INF Maximum Spike Noise Amplitude

Each INF is able to detect and blank a maximum spike amplitude of two volts p-p measured at the input to the INF.

1-8 INF Blanker Switch Trigger

The blanker switch trigger, upon receipt of a spike-detected signal from the detector, provides a control signal to open the blanker switch for the respective RF channel.

1-9 INF Blank Period

Upon detection of a spike, the detector provides a signal to command the in line switch blanker function to open the RF signal channel. The blanker switch is held open for a minimum of one microsecond to a maximum equal to the duration of the detected spike plus one microsecond.

1-10 INF Residual Spikes

There are no residual spikes from the TNS circuits that exceed two times the peak ambient noise floor observed at the TPS receiver A/D converter (the Fast Receiver quadrature A/D inputs for the UFI system) as a result of the blanking process.

1-11 INF Blanker Enable

A manual switch is provided to disable all INFs from opening the blanking switch in response to spike noise.

1-12 Preamp Bias

The TNF provides a 15V ±0.5V dc bias voltage to the RF preamp connected to each input channel. This bias is continuously applied and output through the RF coax connection to each preamp. The TNFE is not required to supply a preamp bias; however, the TNFE provides the bias in support of INF modularity considerations and ease of manufacturing.

1-13 Incident Tracking and Display

The Incident Tracking function provides the logic to increment a counter for each spike detected during the Data in Window signal.

1-14 Count Incrimination

The counter increments one count on each spike detected and blanked by any channel INF during a logic high Receiver Data In Window (DIW) and logic high Receiver Unblank, or a logic high UFI Fast Receiver DIW and logic high Receiver Unblank. The RF Receiver DIW and Receiver Unblank signals are provided by RF Receiver 0 as an active high TTL signal through a series 1kΩ resistor. DIW is a logic high during active data acquisition and Receiver Unblank is a logic low at the end of scan activity. The UFI Fast Receiver Data in Window is provided as an active high TTL signal through a series 1kΩ resistor. Simultaneous blanking by more than one INF increments the counter by one count.

1-15 Local Display

A local display, to indicate the incident counter value, is provided that has at least four digits plus an overload indicator. The display may be disabled by a local switch to save power, part life and to reduce the probability of the radiation of undesirable noise that could degrade image quality during active scanning.

1-16 Counter Power-up Reset

The counter is reset to a value of 0 at time of power-up.

1-17 Manual Counter Reset

The counter has a local, manually activated, reset button that, when activated, resets the count to 0 and clears the overload indication.

1-18 Communications Control and RS-232 Interface Function

The TNS provides communications for the remote reporting of status over an RS-232 interface.

1-19 RS-232 Port Commands

The TNF receives and responds to the following commands:

Read the Incident Counter: An 8-bit word that causes the TNF to transmit back over the RS-232 port a repeat of the command word, followed by an 8-bit word stream formatted to represent the incident counter contents.

Reset the Incident Counter: An 8-bit word that causes the Incident counter to be reset. The TNF repeats the command back to the host over the RS-232 port to confirm the execution of the counter reset.

1-20 Command Response Time

The TNS responds to the RS-232 port commands within one millisecond after receipt of the command, and if the DIW is in the logic low state, or the TNS responds within one millisecond after the next occurrence of a logic low DIW if the DIW was high at time the command was received.

1-21 Response Period

No execution of, or response to, commands are made during a logic high data in window signal.

1-22 Spike Noise Test Function

The TNF provides a Spike Noise Test function to confirm proper operation of the TNS in the presence of a spike noise source. This function introduces spike noise into the input of each INF under test at a level that causes the TNS to actively blank the spikes when the blanker trigger is enabled. The Spike Noise Test circuit is manually connected to any RF channel input to the TNS via a coax cable with BNC connector at time of test only. The circuit is enabled only when in use for testing the desired channel.

1-23 Systems Cabinet Overtemperature Control

The TNS assembly includes the redesign of the existing systems cabinet Overtemperature Control (OTC) function. This OTC function is modular and independent of the TNF and TNFE functions. It is removable for future applications. The systems cabinet OTC monitors the state of the thermal-sensitive switch that is provided with the cabinet. The thermal-sensitive switch closes when the systems cabinet temperature exceeds nominal 94 degrees F.

1-24 Overtemperature Alarm

The OTC provides an audible alarm, upon sensing a closure of the systems cabinet thermal-sensitive switch. The audible alarm emits a pulsating or continuous tone, and is activated for the duration of the closure of the thermal-sensitive switch, or until removal of power from the systems cabinet. The alarm stops if the thermal-sensitive switch opens.

1-25 System Shutdown and Delay Timer

Upon sensing the closure of the thermal sensitive switch, the OTC activates a six-minute ±1-minute delay timer. At completion of the timed delay, the OTC provides closure of a shutdown switch to apply +24 Vdc to the PDU shunt trip breaker. Closure of the shutdown switch activates complete power shutdown for the MRI system. The shutdown switch can handle 24 Vdc at 6 amps maximum contact current. It is normally open when no power is applied to the TNS. If the thermal-sensitive switch opens before timeout of the delay timer, the timer is reset and deactivated. A spare shutdown switch contact is provided for future use.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 2, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.